LAB-ASSIGNMENT : 1
(2401010020_sneha_gupta)

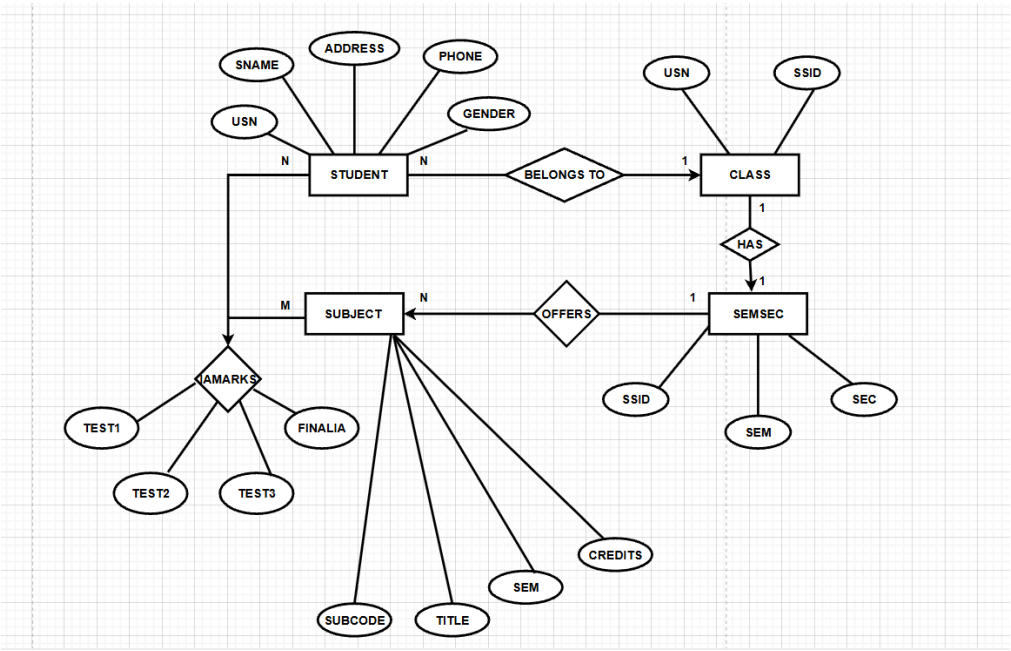
Objective:

To design ER diagrams and convert entities and relationships into relational tables using proper schema notation.

Relational Schema

PART A – COLLEGE DATABASE

STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)



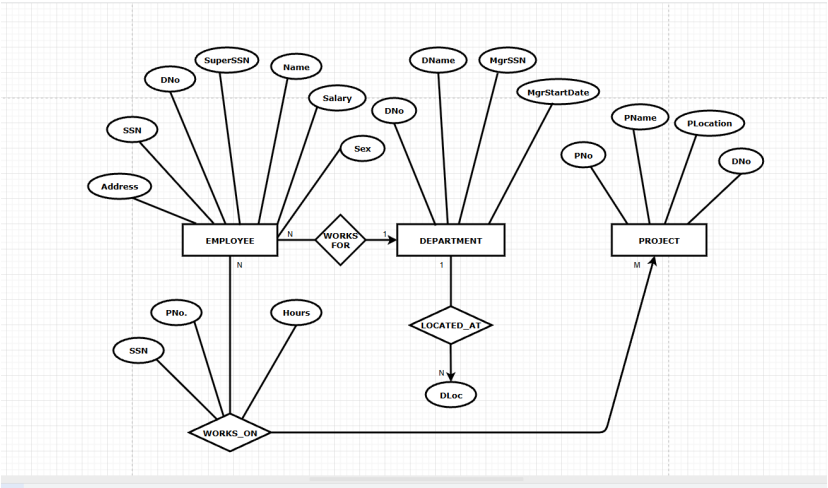
TABLES :

```
mysql> CREATE DATABASE DBMS_SNEHA;
Query OK, 1 row affected (0.09 sec)

mysql> USE DBMS_SNEHA;
Database changed
mysql> CREATE TABLE STUDENT (
    ->     USN VARCHAR(15) PRIMARY KEY,
    ->     SName VARCHAR(50),
    ->     Address VARCHAR(100),
    ->     Phone VARCHAR(15),
    ->     Gender VARCHAR(10)
    -> );
Query OK, 0 rows affected (0.11 sec)

mysql> CREATE TABLE SEMSEC (
    ->     SSID INT PRIMARY KEY,
    ->     Sem INT,
    ->     Sec CHAR(1)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE CLASS (
    ->     USN VARCHAR(15),
    ->     SSID INT,
    ->     PRIMARY KEY (USN, SSID),
    ->     FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    ->     FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
    -> );
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE SUBJECT (
    ->     Subcode VARCHAR(10) PRIMARY KEY,
    ->     Title VARCHAR(50),
    ->     Sem INT,
    ->     Credits INT
    -> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> CREATE TABLE IAMARKS (
    ->     USN VARCHAR(15),
    ->     Subcode VARCHAR(10),
    ->     SSID INT,
    ->     Test1 INT,
    ->     Test2 INT,
    ->     Test3 INT,
    ->     FinalIA INT,
    ->     PRIMARY KEY (USN, Subcode),
    ->     FOREIGN KEY (USN) REFERENCES STUDENT(USN),
    ->     FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
    ->     FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
    -> );
Query OK, 0 rows affected (0.08 sec)
```

PART B – COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Sex, Salary, Superssn, Dno)
DEPARTMENT (Dno, Dname, Mgrssn, Mgrstartdate)
DLOCATION (Dno, Dloc)
PROJECT (PNo, PName, PLocation, DNo)
WORKS_ON (SSN, PNo, Hours)



TABLES :

```
mysql> CREATE TABLE DEPARTMENT (
    ->     Dno INT PRIMARY KEY,
    ->     Dname VARCHAR(50),
    ->     Mgrssn VARCHAR(15),
    ->     Mgrstartdate DATE
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE EMPLOYEE (
    ->     SSN VARCHAR(15) PRIMARY KEY,
    ->     Name VARCHAR(50),
    ->     Address VARCHAR(100),
    ->     Sex CHAR(1),
    ->     Salary INT NOT NULL,
    ->     Superssn VARCHAR(15),
    ->     Dno INT,
    ->     FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
    -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> CREATE TABLE DLOCATION (
    ->     Dno INT,
    ->     Dloc VARCHAR(50),
    ->     PRIMARY KEY (Dno, Dloc),
    ->     FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE PROJECT (
    ->     PNo INT PRIMARY KEY,
    ->     PName VARCHAR(50),
    ->     PLocation VARCHAR(50),
    ->     DNo INT,
    ->     FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

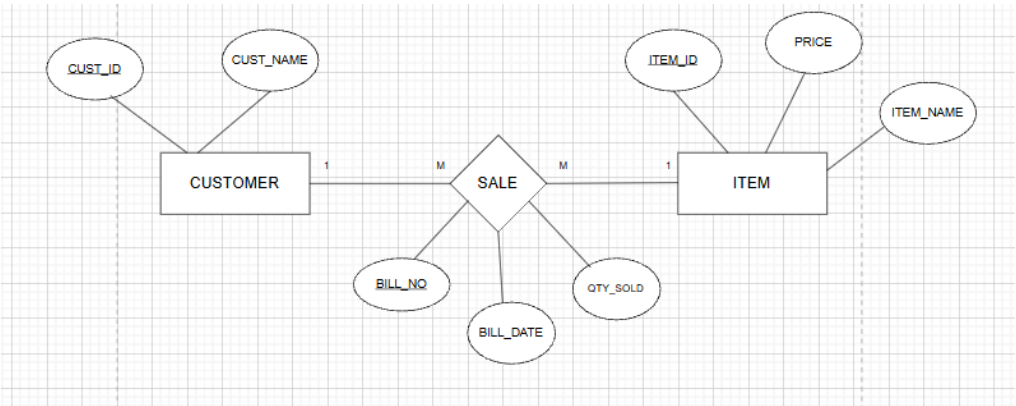
```
mysql> CREATE TABLE WORKS_ON (
    ->     SSN VARCHAR(15),
    ->     PNo INT,
    ->     Hours INT,
    ->     PRIMARY KEY (SSN, PNo),
    ->     FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
    ->     FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

PART C – CUSTOMER SALE SCHEMA

Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name:string, price: integer)
Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)



TABLES:

```
mysql> CREATE TABLE CUSTOMER (
    ->     Cust_id INT PRIMARY KEY,
    ->     Cust_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE ITEM (
    ->     Item_id INT PRIMARY KEY,
    ->     Item_name VARCHAR(50),
    ->     Price INT NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE SALE (
    ->     Bill_no INT PRIMARY KEY,
    ->     Bill_date DATE,
    ->     Cust_id INT,
    ->     Item_id INT,
    ->     Qty_sold INT,
    ->     FOREIGN KEY (Cust_id) REFERENCES CUSTOMER(Cust_id),
    ->     FOREIGN KEY (Item_id) REFERENCES ITEM(Item_id)
    -> );
Query OK, 0 rows affected (0.06 sec)
```

DDL Questions (Data Definition Language):

1. Write SQL statements to create all the above relations with appropriate PRIMARY KEY and FOREIGN KEY constraints.
   DONE!
2. Add a new column Email to the STUDENT relation.

```
mysql> ALTER TABLE STUDENT ADD Email VARCHAR(50);
Query OK, 0 rows affected (0.12 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

3. Apply NOT NULL constraints to Salary and Price attributes.
   DONE!

4. Rename attribute SName to StudentName in STUDENT relation.

```
mysql> ALTER TABLE STUDENT CHANGE SName StudentName VARCHAR(50);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

5. Drop the DLOCATION relation.

```
mysql> DROP TABLE DLOCATION;
Query OK, 0 rows affected (0.06 sec)
```

DML Questions (Data Manipulation Language):

- Insert at least 10 tuples into CUSTOMER, ITEM, and SALE relations.

```
mysql> INSERT INTO CUSTOMER VALUES
    -> (1,'Aman'),(2,'Bhavna'),(3,'Chirag'),(4,'Diya'),(5,'Eshan'),
    -> (6,'Farah'),(7,'Gaurav'),(8,'Hina'),(9,'Ishan'),(10,'Jiya');
Query OK, 10 rows affected (0.03 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO ITEM VALUES
    -> (101,'Pen',10),(102,'Book',250),(103,'Bag',500),
    -> (104,'Bottle',150),(105,'Shoes',1200),
    -> (106,'Notebook',80),(107,'Watch',900),
    -> (108,'Pencil',5),(109,'Box',60),(110,'Laptop',45000);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO SALE VALUES
    -> (1,CURDATE(),1,101,2),
    -> (2,CURDATE(),2,102,1),
    -> (3,CURDATE(),3,103,1),
    -> (4,CURDATE(),4,104,3),
    -> (5,CURDATE(),5,105,1),
    -> (6,CURDATE(),6,106,4),
    -> (7,CURDATE(),7,107,1),
    -> (8,CURDATE(),8,108,10),
    -> (9,CURDATE(),9,109,2),
    -> (10,CURDATE(),10,110,1);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

- List all bills of the current date with customer names and item names.

```
mysql> SELECT S.Bill_no, C.Cust_name, I.Item_name
    -> FROM SALE S
    -> JOIN CUSTOMER C ON S.Cust_id=C.Cust_id
    -> JOIN ITEM I ON S.Item_id=I.Item_id
    -> WHERE S.Bill_date=CURDATE();
+---------+-----------+-----------+
| Bill_no | Cust_name | Item_name |
+---------+-----------+-----------+
|       1 | Aman      | Pen       |
|       2 | Bhavna    | Book      |
|       3 | Chirag    | Bag       |
|       4 | Diya      | Bottle    |
|       5 | Eshan     | Shoes     |
|       6 | Farah     | Notebook  |
|       7 | Gaurav    | Watch     |
|       8 | Hina      | Pencil    |
|       9 | Ishan     | Box       |
|      10 | Jiya      | Laptop    |
+---------+-----------+-----------+
10 rows in set (0.02 sec)
```

- Display total bill amount using quantity and price.

```
mysql> SELECT Bill_no, Qty_sold, Price,
    -> (Qty_sold*Price) AS Amount
    -> FROM SALE S
    -> JOIN ITEM I ON S.Item_id=I.Item_id;
+---------+----------+-------+--------+
| Bill_no | Qty_sold | Price | Amount |
+---------+----------+-------+--------+
|       1 |        2 |    10 |     20 |
|       2 |        1 |   250 |    250 |
|       3 |        1 |   500 |    500 |
|       4 |        3 |   150 |    450 |
|       5 |        1 |  1200 |   1200 |
|       6 |        4 |    80 |    320 |
|       7 |        1 |   900 |    900 |
|       8 |       10 |     5 |     50 |
|       9 |        2 |    60 |    120 |
|      10 |        1 | 45000 |  45000 |
+---------+----------+-------+--------+
10 rows in set (0.00 sec)
```

- List customers who purchased products priced above 200.

```
mysql> SELECT DISTINCT C.Cust_name
    -> FROM SALE S
    -> JOIN CUSTOMER C ON S.Cust_id=C.Cust_id
    -> JOIN ITEM I ON S.Item_id=I.Item_id
    -> WHERE I.Price>200;
+-----------+
| Cust_name |
+-----------+
| Bhavna    |
| Chirag    |
| Eshan     |
| Gaurav    |
| Jiya      |
+-----------+
5 rows in set (0.01 sec)
```

- Count the number of products purchased by each customer.

```
mysql> SELECT Cust_id, COUNT(Item_id) AS Total_Items
    -> FROM SALE
    -> GROUP BY Cust_id;
+---------+-------------+
| Cust_id | Total_Items |
+---------+-------------+
|       1 |           1 |
|       2 |           1 |
|       3 |           1 |
|       4 |           1 |
|       5 |           1 |
|       6 |           1 |
|       7 |           1 |
|       8 |           1 |
|       9 |           1 |
|      10 |           1 |
+---------+-------------+
10 rows in set (0.02 sec)
```

- List products purchased by customer having Cust_id = 5.

```
mysql> SELECT I.Item_name
    -> FROM SALE S
    -> JOIN ITEM I ON S.Item_id=I.Item_id
    -> WHERE S.Cust_id=5;
+-----------+
| Item_name |
+-----------+
| Shoes     |
+-----------+
1 row in set (0.00 sec)
```

- Create a VIEW showing bill_no, bill_date, cust_id, item_id, price, qty,sold, amount.

```
mysql> CREATE VIEW Bill_Details AS
    -> SELECT S.Bill_no, S.Bill_date, S.Cust_id, S.Item_id,
    -> I.Price, S.Qty_sold,
    -> (S.Qty_sold*I.Price) AS Amount
    -> FROM SALE S
    -> JOIN ITEM I ON S.Item_id=I.Item_id;
Query OK, 0 rows affected (0.04 sec)
```

- Create a VIEW showing daily sales date-wise for the last one week.

```
mysql> CREATE VIEW Weekly_Sales AS
    -> SELECT Bill_date, SUM(Qty_sold*Price) AS Total_Sales
    -> FROM SALE S
    -> JOIN ITEM I ON S.Item_id=I.Item_id
    -> WHERE Bill_date >= CURDATE() - INTERVAL 7 DAY
    -> GROUP BY Bill_date;
Query OK, 0 rows affected (0.01 sec)
```